

PRASHANT GAUTAM

Civil Engineer | Post-Fire Assessment, Structural Diagnostics & Resilience

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PROFESSIONAL SUMMARY

Licensed civil engineer with specialized expertise in post-fire damage assessment, non-destructive testing (NDT), and structural diagnostics. Field-experienced in forensic investigations, material testing, ETABS analysis, and developing repair/retrofit recommendations. Seeking graduate research opportunities to advance post-fire assessment methods, material resilience, and rehabilitation strategies.

EDUCATION

Bachelor's in Civil Engineering

2021 - 2025

Pulchowk Campus, Institute of Engineering (IOE), Tribhuvan University (TU)

- Regular program scholarship recipient.

+2 Science

2018 - 2020

Ambition Academy Secondary School, National Examination Board (NEB)

- Recipient of scholarship provided by Kathmandu Metropolitan City.

RESEARCH INTERESTS

- **Post-Fire Assessment & Forensic Engineering:** Damage classification protocols, thermal-induced material degradation, residual capacity evaluation, and forensic reporting to guide repair vs. demolition decisions.
- **Structural Diagnostics & Material Behavior:** Non-destructive testing, destructive sampling, concrete carbonation analysis, and quantification of fire-induced changes in concrete, steel, and masonry properties.
- **Infrastructure Resilience & Rehabilitation:** Retrofit design, repair methodology, cost-effective rehabilitation strategies, and long-term durability of fire-damaged structures.

RELEVANT EXPERIENCE

Civil Engineer (Structural Assessment)

Oct 2025 – May 2026

Institute of Engineering Consultancy Service (IOECS)

- Led post-fire detailed damage assessments (DDA) on 44 federal and central government buildings across Kathmandu and Pokhara.
- Executed concrete core cutting, carbonation depth analysis, rebound hammer testing, and UPV scanning to assess residual material strength.
- Performed ETABS structural analysis to evaluate fire-damaged building capacity and support retrofit recommendations.
- Prepared forensic reports correlating field diagnostics (NDT results, structural observations, material testing) to justify repair vs. demolition decisions.
- Developed comprehensive cost estimates and BOQ for rehabilitation works per Government of Nepal norms.

Civil Engineer (Site Supervision & Measurement)

Aug 2025 – Present

Dibyashwori Nirman Sewa and various JVs linked.

- Supervised site construction to ensure compliance with structural drawings, Nepal National Building Code (NBC), and design standards.
- Led reinforcement inspections (bar bending schedules, placement verification) and formwork stability evaluation.
- Conducted concrete workability tests, material quality checks, and routine inspections prior to structural pours.
- Performed precise quantity take-offs and maintained comprehensive engineering data logs for structural verification.

- Managed site layouts, surveys, and material tracking on active infrastructure projects.
- Oversaw reinforcement inspections and quantity take-offs to ensure efficient material use within budget.

CORE COMPETENCIES

- **Post-Fire Assessment:** Damage assessment protocols, damage classification, thermal analysis, residual capacity evaluation, forensic reporting
- **NDT & Material Testing:** UPV, Rebound Hammer, core cutting, carbonation depth analysis, concrete quality evaluation, strength diagnostics
- **Structural Analysis:** ETABS modeling, thermal effects analysis, RC design (IS 456, IS 1893), capacity evaluation, retrofit design
- **Software:** ETABS, AutoCAD, SW-DTM, Excel (advanced)
- **Design Standards:** IS 456 (Concrete), IS 1893 (Seismic), IS 13920 (Ductile Detailing), NBC (Nepal National Building Code)

ACADEMIC CAPSTONE PROJECT

Seismic Resistant Design of Multi-Storied RCC Commercial Building

2024–2025

- Performed comprehensive structural modeling and dynamic analysis using ETABS for B+G+9 storied building.
- Designed structural members per IS 1893:2016 and IS 13920 for seismic performance optimization.
- Conducted equivalent static and response spectrum analysis per Nepal National Building Code.

PROFESSIONAL DEVELOPMENT & LEADERSHIP

- **Technical Certification:** ADB-sponsored training on Road Slope Stability and Bioengineering (RCIP-AF TA-10113, Nepal).
- **Team Leadership:** Project group leader; Academic Class Representative.
- **Magazine Editor:** Coordinator and editor, 'The Bridge' Magazine (CESS-Nepal).
- **Exhibition Organizer:** Primary coordinator, campus health camp; core technical coordinator, 'AADHAR-2081' Civil Engineering Exhibition.